

Calculators, cell phones, or notes are not allowed.

- (1) Compute all the elements of $T(11) \subset F_{11}[\sqrt{2}]$ and find the generators of $T(11)$.

$T(11) = \{\pm 1, \pm 4\sqrt{2}, \pm 3 \pm 2\sqrt{2}, \pm 6 \pm \sqrt{2}\}$ (the point is that $1 = 1^2 - 2 \cdot 0^2 = 0^2 - 2 \cdot 4^2 = 3^2 - 2 \cdot 2^2 = 6^2 - 2 \cdot 1^2$). $(-3 \pm 2\sqrt{2})^k$ for $k = 1, 5, 7, 11$ are the 4 generators. In fact $(-3 + 2\sqrt{2})^2 = 6 - \sqrt{2}$, $(-3 + 2\sqrt{2})^4 = 5 - \sqrt{2} \neq 1$, $(3 + 2\sqrt{2})^6 = (6 - \sqrt{2})(5 - \sqrt{2}) = 32 - 11\sqrt{2} = -1 \neq 1$. So the order of $-3 + 2\sqrt{2}$ divides 12 and doesn't divide 4 or 6. Thus the order of $-3 + 2\sqrt{2}$ is 12.

- (2) Show that if $z = x + y\sqrt{d} \in F_p[\sqrt{d}]$ where d is not a square mod p , then $z^p = x - y\sqrt{d}$.

$z^p = (x + y\sqrt{d})^p = x^p + y^p \sqrt{d}^p = x + y\sqrt{d} \cdot d^{(p-1)/2} = x - y\sqrt{d}$. (We have used FLT to conclude $x^p = x$ and $y^p = y$, the fact that in characteristic p we have $(a+b)^p = a^p + b^p$ and the fact that $d^{(p-1)/2} = -1$ since d is not a square mod p).